

PROJECT PROFILE

FOR

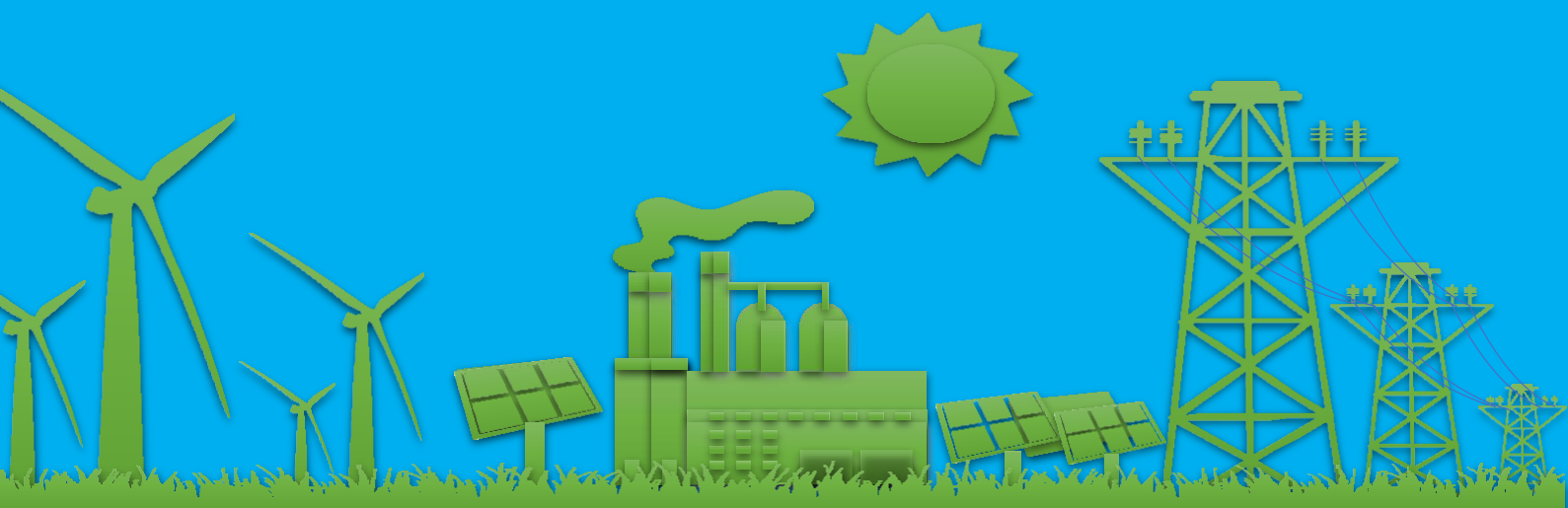
PLASTIC WASTE

Disposal & Energy
Recovery

By



T & M industries pvt.ltd



MAKE IN INDIA



&

SWATCH BHARAT
INITIATIVE



Plant Machinery Design, Manufacturing & R&D
Done Under the Banner of
MAKE IN INDIA
Initiative

A Dedicated Research & Development

*To achieve target of **PLASTIC** and **POLLUTION FREE** India*



This Project Will Help Achieve the Target of Clean and Pollution Free India with Sustainable Development Model & State of Art Technologically Advanced Machinery.



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(1) MUNICIPAL SOLID WASTE

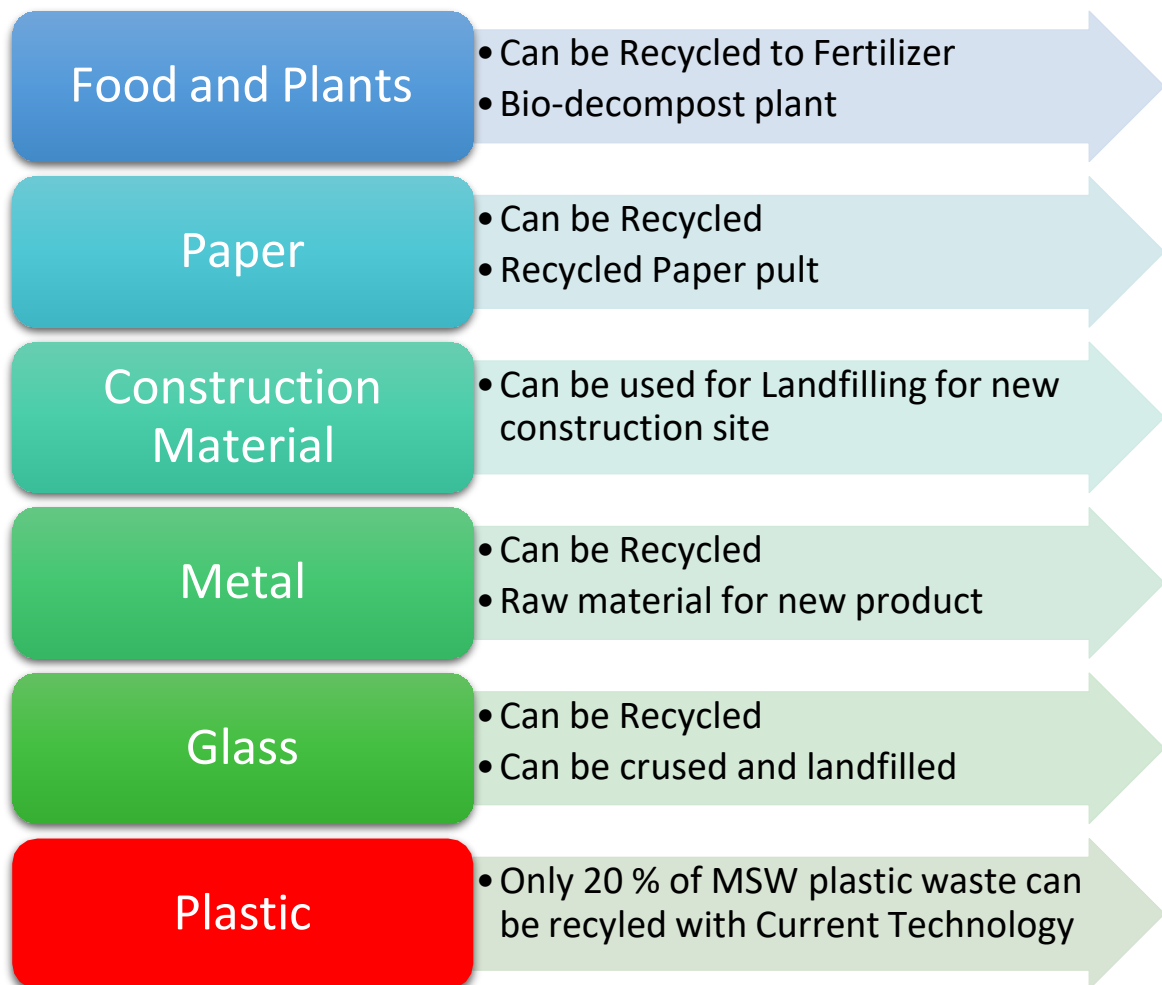
1.1 Type of waste and data

Municipal solid waste contains all type of mixed garbage like Wood, plastic, glass, cloths, biodegradable material, metals bricks and dirt and such products as packaging, grass clippings, furniture, clothing, bottles and cans, food scraps, newspapers, appliances, consumer electronics, and batteries. These wastes come from homes; institutions such as schools and hospitals; and commercial sources such as restaurants and small businesses.





- The most of Municipal solid waste can be recycled.



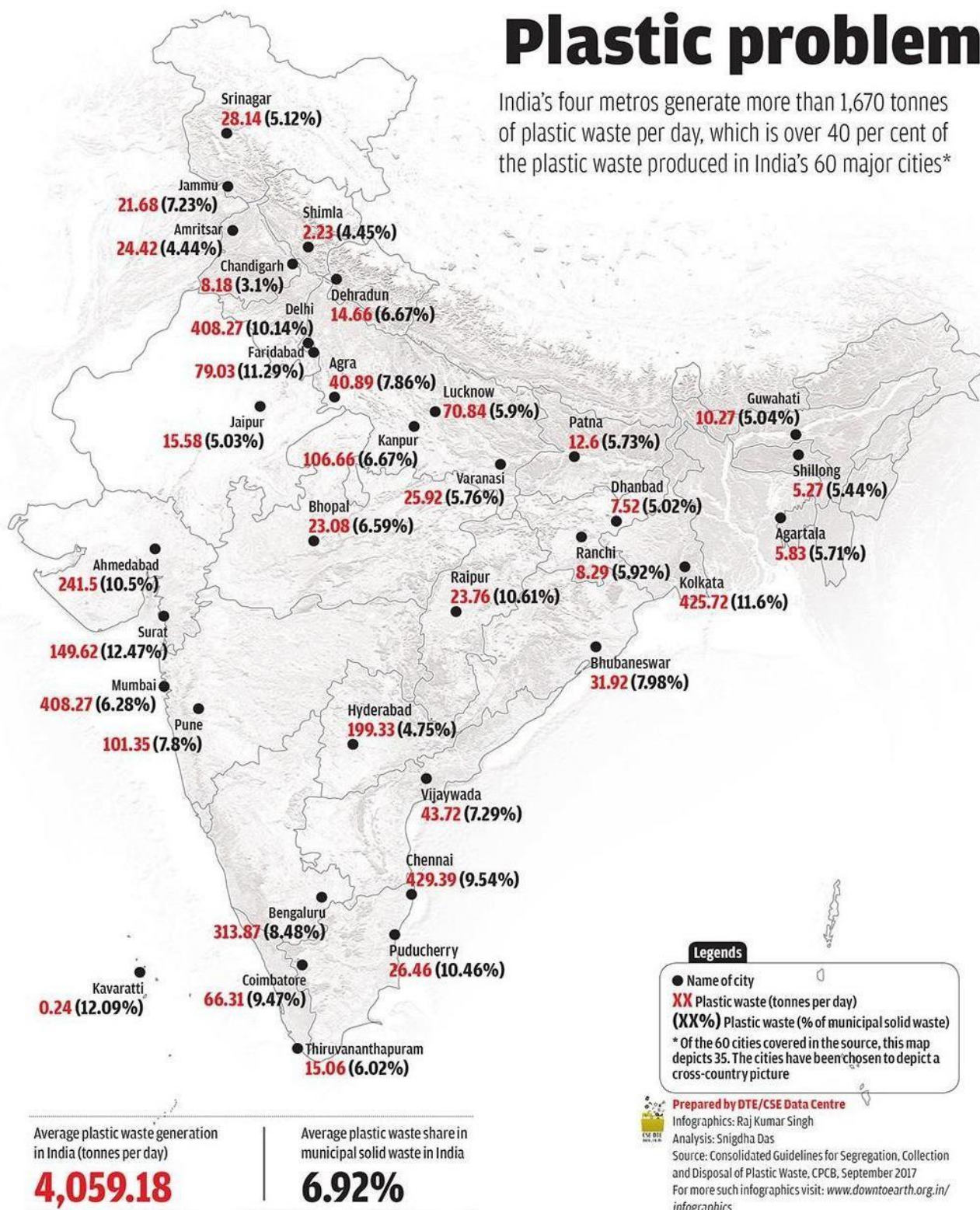
WHAT ABOUT REST OF 80% non-recyclable plastic waste in MSW???

This 80 % directly go to the LANDFILLING or Burnt in open environment, which pollutes air and land with **irreversible effects**. Like toxic air pollutants causing asthma and Cancer for Human. And Land become unfertile which effects ground water and pollutes it.

The plastic stay in Landfill for **200 to 400 Years**.

1.2 Plastic waste data

Below Figure shows the generation of Municipal Solid waste (MSW) from each major city. Data in **RED numbers** shows plastic waste generation in TONNES per DAY.





1.3 Plastic waste problem and cause

Researchers estimate that more than 8.3 billion tonnes of plastic have been produced since the early 1950s. About **60%** of that plastic has ended up in either a landfill or the natural environment.

Plastic pollution involves the accumulation of plastic products in the environment that adversely affects wildlife, wildlife habitat, or humans. Plastics as pollutants are categorized into micro, meso or macro debris based on size. Plastic pollution can unfavourably affect lands, waterways, oceans, living organisms and marine animals can also be affected through direct ingestion of plastic waste.

Effects of Plastic Pollution

It seems rather obvious that this amount of a material that isn't meant to break d wreak havoc on natural environments, leading to long-term issues for plants, people. Some of the major long-term effects of plastic pollution are:

It Upsets the Food Chain

Because it comes in sizes large and small, polluting plastics even affect the world organisms such as plankton. When these organisms become poisoned due to plastic ingestion, this causes problems for the larger animals that depend on them for cause a whole slew of problems, each step further along the food chain. Plus, it nano plastic are present in the fish that many people eat every day.

Groundwater Pollution

world's water is in great danger because of leaking plastics and waste. If you seen a garbage dump, imagine what happens every time it rains – then imagine that your drinking water. Groundwater and reservoirs are infected by leaking toxins and heavy metal.

Most of the litter and pollution affecting the world's oceans also derives from plastic had terrible consequences on many marine species, which can lead to consequences those that eat fish and marine life for nutrients – including people.



Land Pollution

When plastic is dumped in landfills, it interacts with water and form hazardous chemical. When these chemicals seep underground, they degrade the water quality. Wind can deposit plastic from one place to another, increasing the land litter. Animals that may come in Contact with plastic might suffocate them to death.

Air Pollution

Burning of plastic in the open air, leads to environmental pollution due to the release poisonous chemicals. The polluted air when inhaled by humans and animals affect health and can cause respiratory problems like cancer and asthma.

It is Poisonous

Man, artificially makes plastic by using a number of toxic chemicals. Therefore, us exposure to plastics has been linked to a number of health concerns affecting people of the world. The processes of making, storing, disposing of, and just being around be extremely harmful to living things.

It is Expensive

It costs millions of rupees each year to clean affected areas after exposure, not to the loss of life to plants, animals, and people. As land becomes more valuable, place to put garbage is becoming a problem in many parts of the world. Plus, excess pollution has led to decreased tourism in affected areas, significant those economies.

We need to slow the flow of plastic at its source, but we also **need to improve the way we manage our plastic waste**. Because right now, a lot of it ends up in the environment.

1.4 Recently used Plastic waste recycling technologies

1.4.1 MSW DUMP SITE

Currently widely used method for **dumping Municipal Solid waste**.

The waste is piled over a large area, in multi-layer order on open wide ground.



Problems occurs due to this Method

- The most adverse effect of this method is **LOST** of Fertile ground, where we can **never** do FARMING or **never** able to do Tree plantation again.
- The dump site requires **larger area** which is a disadvantage for Large cities, Because the Major cities required large area for habitation but the MSW dump sites grab very massive area.
- The most concerning problem of landfills is the **toxic waste** in the pileup will blend with the water, when it rains and that can flow out of the landfill, contaminate ground water, damage crops and can cause serious health problems.
- Decomposition of waste **produces Methane**. Methane is fatal for humans and can cause significant damage to our eco-system. Methane can cause chronic and terminal diseases. It can also light up easily and entire landfills can be on fire in no time. Methane also increase **GLOBAL WARMING**.
- If recyclable materials are put into landfill sites you use more energy and resources trying to create new materials instead of re-using or recycling so landfill sites should not be the first port of call for re-usable or recyclable materials.

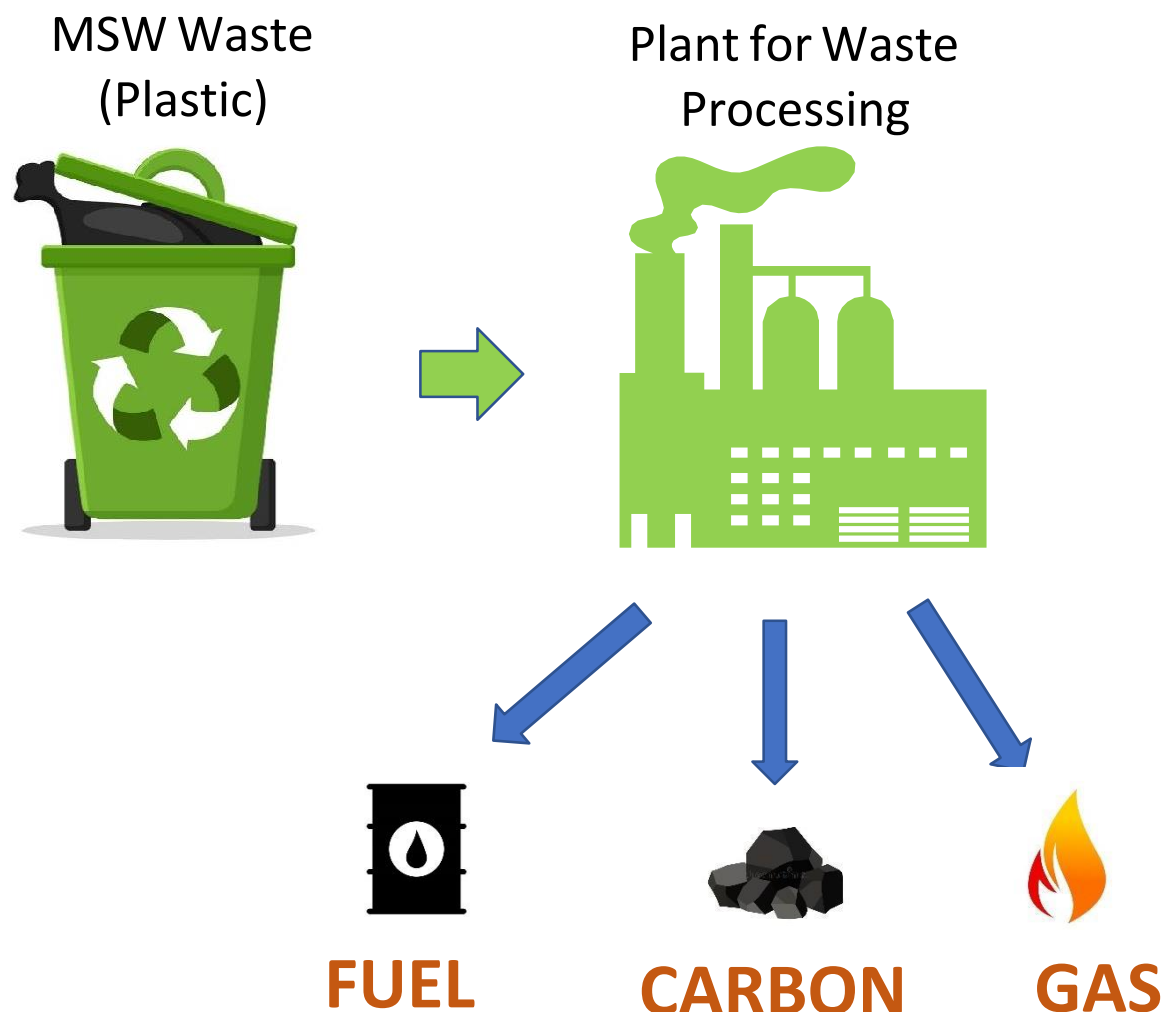
1.4.2 PLASTIC BURNING



- It is second largely and most **Dangerous** used method to get rid of plastic waste at MSW dump sites.
- The smoke produced includes acid gases, **carcinogen** dioxin, particulates, heavy metals, and nitrogen oxide. These gases **are poisonous** to the environment. Research has shown that dioxin produced is a cancer forming chemical.
- Even though the ash that remains from the process can be comparatively small in quantity, it contains a number of **poisons** and heavy metals which requires further treatment. If not disposed correctly, it can **cause serious harm** to the public and the environment.
- Burning of plastic does not encourage recycling and waste reduction. This is not a calculated strategy for any society. The point of focus should be on reducing waste and recycling most of it. Merely burning most of the waste without recycling some of it will only further environmental damage because it may encourage more waste production.

We have Solution.

Waste to Energy

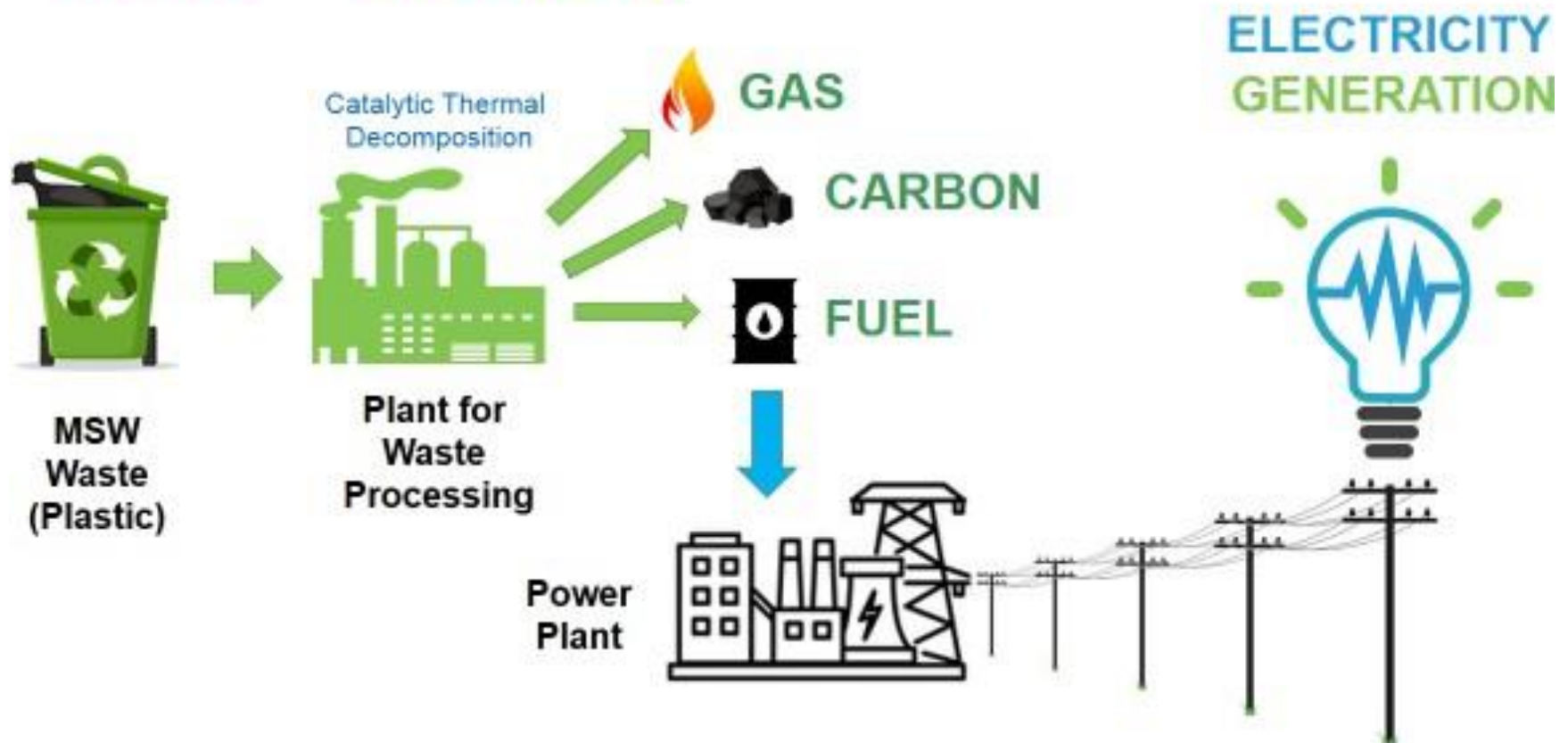


Our Plant Recycling Capacity
6000 kg waste plastic per DAY



Countries using Waste to energy Process for Generation of Electricity from Waste management Process

Waste to Electricity



1.5 Government laws



सत्यमेव जयते

GOVERNMENT OF INDIA
MINISTRY OF URBAN DEVELOPMENT
<http://moud.gov.in>

MUNICIPAL SOLID WASTE MANAGEMENT MANUAL

Central Public Health and Environmental
Engineering Organisation (CPHEEO)

IN COLLABORATION WITH



giz Deutsche Gesellschaft
für Internationale
Zusammenarbeit (giz) GmbH

German International Cooperation

2016

Non-recyclable waste having calorific value of **1500 kcal/kg** or more **shall not be disposed of on landfills** and **shall only be utilised for generating energy** either or through refuse derived fuel or by giving away as feed stock for preparing refuse derived fuel.

3.5.1 LEGAL FRAMEWORK FOR REFUSE DERIVED FUEL PRODUCTION AND UTILISATION



As per SWM Rules, 2016:

Clause 15: Duties and responsibilities of local authorities:-

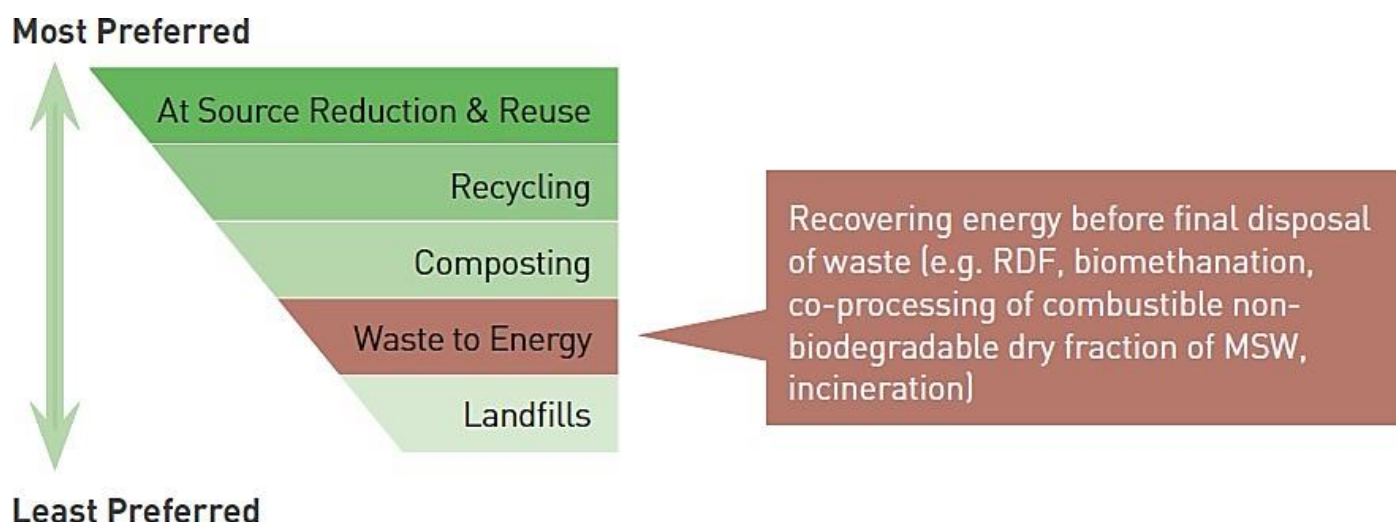
(v) facilitate construction, operation and maintenance of solid waste processing facilities and associated infrastructure on their own or with private sector participation or through any agency for optimum utilisation of various components of solid waste adopting suitable technology including the following technologies and adhering to the guidelines issued by the Ministry of Urban Development from time to time and standards prescribed by the Central Pollution Control Board. Preference shall be given to decentralised processing to minimize transportation cost and environmental impacts such as-

b) waste to energy processes including refused derived fuel for combustible fraction of waste or supply as feedstock to solid waste based power plants or cement kilns;

18. Duties of the industrial units located within one hundred km from the refused derived fuel and waste to energy plants based on solid waste- All industrial units using fuel and located within one hundred km from a solid waste based refused derived fuel plant shall make arrangements within six months from the date of notification of these rules to replace at least five percent of their fuel requirement by refused derived fuel so produced.

STATEMENT FROM MANUAL ON MUNICIPAL
SOLID WASTE MANAGMENT
MINISTRI OF URBAN DEVLOPMENT

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Its stated in above figure waste to energy should given more importance than landfill dumping.

It is assumed that at least 65 to 80% of energy content of waste can be recovered as usable form of energy through waste to energy technologies.

(2) CATALYTIC THERMAL DECOMPOSITION

2.1 Recycling Flow chart



- The waste from City is collected by Municipal Waste collection system and stored at MSW DUMPING SITE outside the city area.
- The Plastic waste is separated from other garbage and transported to “WASTE TO FUEL” Plant Facility for further process.
- The Plastic waste then passes through Catalytic Thermal Decomposing process.
- After process Completion, the plastic waste converted to two use full resources: **Fuel oil & Carbon.**
- There is NO OTHER DISCHARGE OUT OF PLANT, that means it is a **ZERO DISCHARGE from plant.** No more plastic pollution.

2.2 Raw material can be used

Plastic type	Source	Figure
PET (Polyethylene terephthalate)	Plastic Bottles for Soda and Water, Microwave Food Trays, Some of Textiles 	
HDPE High density polyethylene PE polyethylene	Plastic Bags, Milk pouch, Jugs, Detergent Bottles, Cosmetic Bottles, Watering cans, Plastic chairs, Plastic bags, Plastic lumber 	
LDPE Low Density Polyethylene	Carry Bags, packing material, Stretch Wrap, Wire Coverings 	
PP Polypropylene	Bottle Caps, Automotive Parts, Appliances, food Wrappers, Computer Parts 	
PS Polystyrene	Cutlery, Cups, CD Cases, Packing, Food Containers, Styrofoam 	

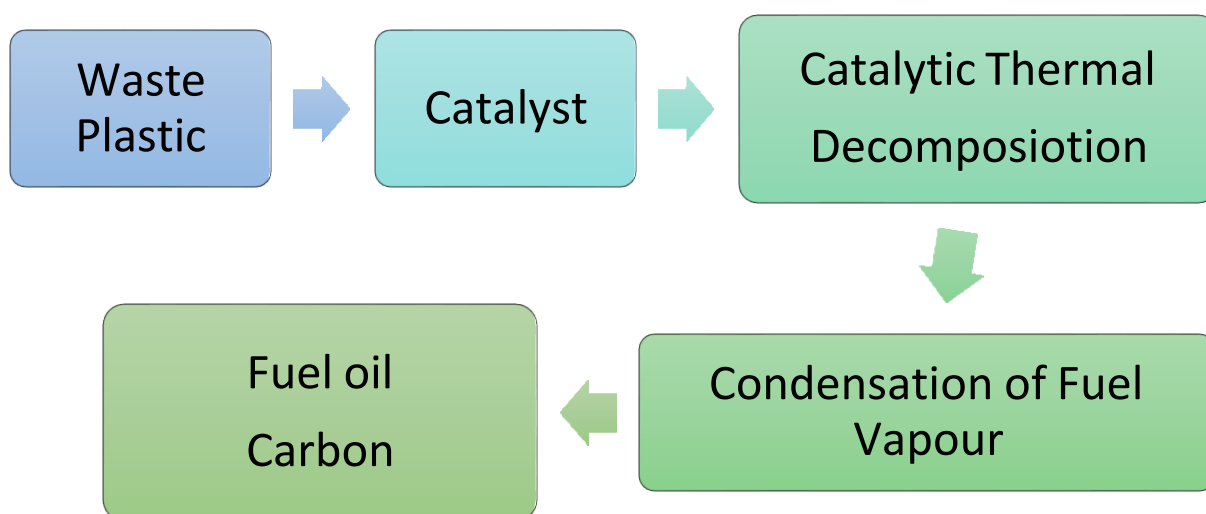
Other Special Type of plastic waste Acceptable for this recycling Plant

- We have a new innovative plant design which can accept type of **Mix Plastic**. That means we can feed all **Mix plastic waste** directly **without separation** the different type of plastic.
- Our Plant is designed to handle **MLP** (Multilayer Plastic) waste. Also acceptable in mix form with other plastic waste.
- Plant design is suitable for Handling **MSW (Municipal Solid Waste) plastic waste**. The robust design of plant can handle MSW plastic waste with ease and obtain good product yield.
- Industrial packaging waste plastic can be used as feed stock and other plastic waste like **ABS** and **Rubber Mixed Plastic waste** is suitable for this Plant Technology.
- Multi-Layer plastic, Plastic Carry Bags, All mix plastic waste is Acceptable for Input **(Rafia, panni, cable cover, etc.)**

Our special Catalyst that has be developed by Intensive Research & Development, able to convert any type of plastic waste into useful product with almost zero WAX content.

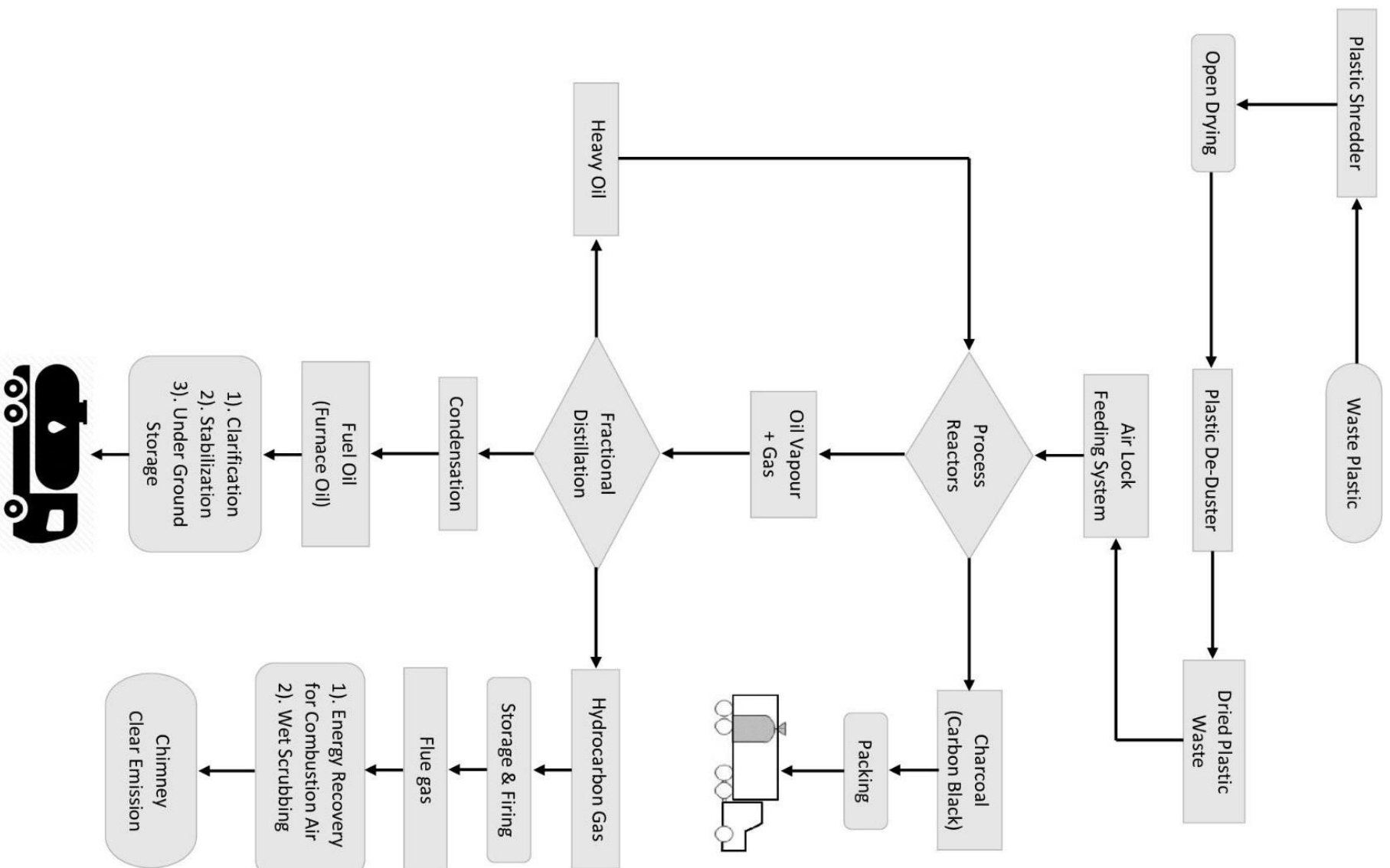
2.3 Processes information

The plastic waste feed in to Continues Process Vessel. The Process Vessel is heated using HAG (HOT AIR GENERATOR) burner system. When the temperature of plastic reaches required temperature, fuel vapour will start generating. Fuel oil vapour produce from the waste plastic **Catalytic Thermal Decomposition** will be cooling down by the condenser, it would be liquefied and go into the oil storage tank. The fuel vapours which can't be liquefied, designed to go back to combustion system and use as BURNER FUEL for HAG system. The non-condensable gas can be utilized to heat the reactor as fuel, which will save energy for the whole working process.



Through the waste plastic **Catalytic Thermal Decomposition** process, **plastic waste** can be **effectively** turned into two useful resources, such as **fuel oil and Carbon**. Both of the end-products can be used as fuels for burning.

Process FLOW CHART



2.4 Product obtained



Fuel



30%



Carbon



35%



Gas



35%

Product uses



FUEL

Used as Industrial
Burner fuel
- LDO (Light Diesel oil)
- FO (Furnace oil)



CARBON

Use as Solid Fuel
In Cement Clines
**Can be Used as
Alternative Fuel for
Industries such as Coal**



GAS

Use as Fuel
in our plant
Heating System
Clean burning Fuel

2.4.1 Fuel Oil

- Fuel Oil Obtained from Catalytic Thermal Decomposition process, can be widely used as fuel for FURNACE, Industrial Boilers, other heating application.
- The Sulphur content of fuel is lower than any Petroleum fuel (Diesel, LDO, Kerosene, etc.) used nowadays.
- Due to lower Sulphur content, it is **ECO FRIENDLY Fuel**.
- The sulphur is main effecting parameter for **ACID RAIN**, but our fuel does **not** produce any sulphur emissions.
- The properties of fuel are good for using it as fuel for Furnace and Industrial Boilers, as has less pollution emission.



2.4.2 Charcoal

- Second Product obtain from this process is pure charcoal which can be used in cement Clines as fuel source.
- The Charcoal can be use as **Alternative fuel** for Wood and Coal.
- This Charcoal has **lower emission** compare to coal used in industries nowadays.
- It also has Lower Sulphur Contain compared to Normal Coal.



2.5 ZERO WASTE DISCHARGE PLANT

- As we Discussed above the Catalytic Thermal Decomposition process converts waste plastic waste in to useful resources like Fuel oil and Carbon.
- There is **no** other Bi-product generated during this process.
- That means this Catalytic Thermal Decomposition Process **REMOVES (Destroys)** the PLASTIC waste from **ENVIROMENT**.
- So, this process is a ZERO WASTE DICHARGE, which Actually converts plastic waste to all usable resource without any unwanted Left-over material.
- It ENERGY POSITIVE PLANT, that means plant generates more energy than it consumes.
So, we **Generate 25 X Times more energy than we consume**.
Hence we have carbon negative foot print.
- **THIS PROCESS IS CLEAN SOURCE OF ENERGY.**

(3) PLASTIC RECYCLING PLANT STATISTICS

3.1 GPCB RULES AND REGULATIONS & PROCESS APPROVED BY CPCB & SWATCH BHARAT MISSION

Design of Plant obeys the ALL GPCB rules and regulation for safety and Pollution.

Plant Design follows the all rules and regulation declared in this GAZETTE OF INDIA.

THE PLASTIC WASTE MANAGEMENT RULES, 2016

रजिस्ट्री सं० डी० एल०-33004/99

REGD. NO. D. L.-33004/99



असाधारण

EXTRAORDINARY

भाग II—खण्ड 3—उप-खण्ड (I)

PART II—Section 3—Sub-section (I)

प्राधिकार से प्रकाशित

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MINISTRY OF ENVIRONMENT, FOREST AND CLIMATE CHANGE

NOTIFICATION

New Delhi, the 18th March, 2016

Data taken from [ANNEXURE-2]

3.2 PROCESS APPROVED BY CPCB & SWATCH BHARAT MISSION

Below Mentioned the procedure of Waste Plastic to Fuel Process.

Conversion of plastic waste into liquid RDF (Oil):

Firstly, plastic waste is segregated mechanically from municipal solid waste (MSW). This method is not full-proof, as along with plastic waste, other lighter material is also segregated. Therefore, the segregated plastic waste is again sent through conveyor belt fixed with optical segregation device for 100% source segregation of plastic waste. Then, the mixed plastic waste is converted into the more useable i.e. Liquid RDF through catalytic pyrolysis. The process is termed random De-polymerization as the degradation of bonds to break down into monomers occurs randomly. This process is utilized for the degradation of plastic to lower hydrocarbons. The random De-polymerization is carried out in a specially designed Reactor, in absence of oxygen and in the presence of certain catalytic additives. The maximum reaction temperature is 350°C. The entire feed material is converted into either of the products: Liquid RDF, gases and some sludge. There is no effluent generated in the process and the unused hot Air from the reactor is released through chimney. Steps involved in conversion of plastic waste into liquid fuel RDF (Fuel) are given below

Mechanical segregation of plastic waste from mixed MSW dump yard/storage; Transportation of segregated plastic waste through conveyor belt for optical segregation;

- Optical segregation of plastic waste (only HD, LD, PP and multilayer packaging's except PVC);
- Shredding of plastic waste and dislodging dust and impurities;
- Transportation of segregated (100% plastic waste) into feeding hopper (reactor);
- Feeding of plastic waste into reactor for random depolymerization in presence of additives;
- Collection of liquid RDF (Fuel);
- Collection of rejects and solid waste (Carbon).

The schematic flow diagram is shown at Figure 3 and process flow diagram is shown at

Data taken from [ANNEXURE–3]

3.3

PROJECT INFORMATION

- This Waste plastic to energy is LATEST TECHNOLOGY FIRST TIME introduce IN INDIA by An INDIAN Company for MSW **plastic waste Disposal**.
- ALL Research & Development work carried out in INDIA by Professionals and Engineer for Optimized Performance of Plant.
- It's India's First **"SWADESI"** PLASTIC WASTE TO ENERGY Plant Design Indigenously Design in India, after studying the requirement for MSW Plastic Waste Disposal.

All made in India plant design and machinery.

Zero Energy Consumption plant

- The Energy Required for Plant Working is Actually Zero.
- This is because of Non- Condensable Fuel Vapour used as **Fuel for Burner System**.
- So, only very Less energy required to start the plant, after a stage Plant required no other energy.
- So, It's a ZERO ENERGY Consumption Plant.

Fully Automatic Plant (PLC controlled)

- Fully Automatic PLC controlled Plant and Safety feature
 - Automatic PLC controlled Plant Design
 - Continues Process
 - Oxygen Sensor
 - Pressure sensor
 - Temperature Sensors
 - Air Lock Feeding System
 - Flow meters
 - Air trap Valves
 - Nitrogen Purging
 - Wet Scrubber
 - Emergency Blow off Vent
 - Fully Automatic PLC Controlled Burner System
 - Hot air generator
 - Air Trap Valve Mechanism
 - Use of Non- Return Valves for more safety
- 
- ✓ State of the art Fully Automatic PLC controlled Plant operates with minimum manual inputs.
 - ✓ Design of Plant obeys the ALL CPCB rules and regulation for safety and Pollution.
 - ✓ Being a hydrocarbon handling project Safety is our 1st priority. Special Design to ensure very full proof operation of reactor.
 - ✓ PLC unit controls the temperature of reactor precisely, design to perfectly decompose waste plastic onto use full energy.
 - ✓ The PLC unit controls Burners of hot air generator for accurately provide required heating for reactor.
 - ✓ Oxygen sensors, temperature sensors, pressure sensors attached to each reactor to ensure top safety.

- ✓ The flow meter and pressure sensor fitted on reactor to monitor any blockage or pressure rise.
- ✓ For Zero pollution of air, State of the Art Column Type Wet scrubber used to remove all impurities from flue gases. (As per CPCB Rules and Regulations)
- ✓ Air lock feeding system and air trap valve prevents any air from entering the reactor chamber. The reactors enclosed inside a hot air chamber, so no direct contact of reactor with atmosphere. Improves heating efficiency and Safety.
- ✓ Control valves used in plant are Fully Automatic, which is controlled by PLC unit.
- ✓ PLC control Module collect data from Temperature sensor, Pressure sensor, Oxygen Sensor in real time and Monitor for any variation or malfunction.
- ✓ Hot air heating for achieving effective and continues control of reactor temperature without direct heating the reactor body. **(No direct heat given to Reactor)**
- ✓ PLC logic unit automatically check all sensors, valves and control system to locate the problem in plant, exactly which part or system have malfunctioned and Display that problem in PLC control Board, so that failed part or system can be replace with new one.
- ✓ For Fire Fighting Purpose in emergency Situation, the Water Sprayer and Fire extinguisher are fitted in all plant Premises which start automatically when any fire detected. (Over Head System)

Employment Generation



It should be noted that this project will generate indirect job creation also.

Employment is an important part of the economic, social and environmental development process and procedure of any country. Employment provides financial freedom and decision making power. Employment opportunities for citizens in India can help to reduce poverty, pollution & upbringing of all class of people of India.

ANNEXURE

1 CPCB DATA ON WASTE MANAGEMENT

<http://cpcb.nic.in/status-of-implementation-of-plastic-waste/>

2 CPCB GUIDELINES FOR MSW MANAGEMENT

<http://cpcb.nic.in/plastic-waste/>

3 SWACHH BHARAT MISSION MUNICIPAL SOLID WASTE MANAGEMENT MANUAL

<http://swachhbharatmission.gov.in/sbmcms/index.htm>





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